W-L-001289(GSA-FSS)

February 28, 1967

# INTERIM FEDERAL SPECIFICATION

## LAMP INCANDESCENT

## (ELECTRIC, MEDICAL INSTRUMENT TUNGSTEN-FILAMENT)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers incandescent lamp of one size for operation on 1.5 volts. This lamp is primarily designed for use with Medical Instruments.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

| | | |
|---|---|---|
| PPP-B-585 | - | Boxes, Wood, Wirebound |
| PPP-B-601 | - | Boxes, Wood, Cleated-plywood |
| PPP-B-621 | - | Boxes, Wood, Nailed and Lock-corner |
| PPP-B-636 | - | Box, Fiberboard |
| PPP-B-640 | - | Boxes, Fiberboard, Corrugated, Triplewall |

Federal Standards:

| | | |
|---|---|---|
| Fed. Std. No. 123 | - | Marking for Domestic Shipment (Civilian Agencies) |
| Fed. Test Method Std. 151 | - | Metals; Test Methods |

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 6240

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Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

**(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)**

2.2 Other publications. The following publications form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

Illuminating Engineering Society:
General Guide to Photometry

(Application for copies should be addressed to the Illuminating Engineering Society; 345 East 47th Street, New York, New York 11117).

3. REQUIREMENTS

3.1 Material.

3.1.1 Brass. Brass used for the lamp shell, base and contact sleeve shall be free-cutting, ASTM B-16, except the brass shall contain not less than 56 percent copper.

3.1.2 Insulator. The insulator shall be hard rubber phenolic plastic, or equal.

3.1.3 Glass. Glass shall be clean and free of bubbles.

3.2 Style and design. Lamps shall conform in all respects to the requirements contained in table I. The base of each lamp shall be such that it shall fit into the receptacle of the applicable instrument indicated in 6.1, when tested as specified in 4.4.1.

Table I

| Volts | Amperes (1) | Bulb (2) | Finish |
|---|---|---|---|
| 1.5 | 0.13 | T-3/4 | Clear |

(1) Tolerance ± 10 percent (see 4.4.1)
(2) T-Tubular, see Fig. 1

3.3 Identifcation marking. The manufacturer's name or registered trademark shall be legibly and permanently marked on the lamp shell.

3.4 Sterilization. The lamp shall be capable of being sterilized as specified in 4.4.2, without impairing its function or appearance.

3.5 Workmanship. Workmanship shall be first class throughout. Incandescent lamps shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. Such examinations and tests as are set forth in this specification or as shall otherwise be appropriate or necessary to insure that the supplies conform to specification requirements, shall be performed by and at the expense of the supplier. Suppliers who do not have facilities addquate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 2 years after delivery of the supplies to which such records relate.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.



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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Military Standard MIL-STD-105.

| | Inspection level | AQL (defects per hundred) |
|---|---|---|
| For visual examination | | |
| Major defects | II | 1.0 |
| Total (major and minor combined) defects | II | 4.0 |
| For dimensional examination | S3 | 2.5 |

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Military Standard MIL-STD-105.

| Characteristic | Requirement | Test Procedure | Inspection Level * |
|---|---|---|---|
| Base fit | 3.2 | 4.4.1 | S2 |
| Electrical properties | 3.2 | 4.4.1 | S2 |
| Resistance to sterilization | 3.4 | 4.4.2 | S2 |
| Spherical candlepower | 3.2 | 4.4.3 | S2 |

*Acceptance number shall be zero, and not less than three samples shall be taken regardless of lot size.

4.2.3 Raw materials. Four ounces of each lot of brass used in the construction of the lamps shall be selected for chemical analysis.

4.2.4 Components. Sufficient samples of the components shall be selected from each lot of material used in the manufacture of the lamps to determine compliance with the requirements of this specification.

4.3 Examination. The lamps shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major
101 Glass envelope not properly secured to base.
102 Lamp does not fit into specified receptacle.
103 Lamp does not light.
104 Glass envelope cracked.

Minor
201 Identification marking missing, incomplete, illegible, or impermanent.
202 Glass envelope not free of bubbles, dirt, and other imperfections.
203 Finish not smooth; not free of burs and cracks.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The lamps shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.



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4.4 Tests. Tests shall be conducted, to determine compliance with specification requirements.

4.4.1 Operation test. The sample lamp shall be installed in the appropriate lamp receptacle to determine proper fit, and shall be operated at the specified voltage to determine that the lamp operates. In addition, the current shall be measured on an instrument accurate to the nearest 0.001 ampere.

4.4.2 Sterilization test. The sample lamp shall be completely immersed in 95 percent ethyl alcohol for not less than 6 hours.

4.4.3 Spherical candlepower. Spherical candlepower shall be determined in accordance with the methods outlined in the General Guide to Photometry. Each sample shall be aged for a period of 1/2 hour by operation at the lowest specified voltage before testing.

4.4.4 Chemical composition. Tests for chemical analysis shall be conducted in accordance with method 111.1 or 112.1 of Federal Test Method Standard No. 151.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table II. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

Table II. Classification of preparation for delivery defects

| Examine | Defects |
|---|---|
| Markings (exterior and interior) | Omitted; incorrect; illegible; improper size, location, sequence, or method of application. |
| Materials | Any component missing or damaged. |
| Workmanship | Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container. |

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Six lamps shall be packaged in a paperboard, plastic or metal container commercially supplied.

5.1.1.2 Intermediate packaging. Twenty-four unit packages of lamps shall be intermediate packaged in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed and waterproofed in accordance with the appendix to the box specification.

5.1.2 Level B.

5.1.2.1 Unit packaging. The lamps shall be unit packaged as specified in 5.1.1.1.

5.1.2.2 Intermediate packaging. Twenty-four unit packages of lamps shall be intermediate packaged in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with method I of the appendix to the box specification.

5.1.3 Level C. The lamps shall be packaged to afford protection against deterioration and damage during shipment from the supplier to the initial destination.



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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. The lamps in quantities as specified (see 6.2) shall be packed in a close-fitting box conforming to PPP-B-585, class 3, PPP-B-601, overseas type, PPP-B-621, class 2, PPP-B-636, class weather-resistant or PPP-B-640, class 2. The boxes shall be closed and strapped in accordance with the appendix to the applicable specification. The gross weight of the PPP-B-636 box shall not exceed the weight limitations of the box specification. When the gross weight of the wood or triple wall boxes exceeds 250 pounds, skids shall be provided in accordance with the applicable box specification.

5.2.2 Level B. Four intermediate packages of lamps shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with method I of the appendix to the box specification.

5.2.3 Level C. The lamps shall be packed to insure carrier acceptance and safe delivery at destination, in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lamp furnished under this specification is for use on various bronchoscopes, esophagoscopes, and laryngoscopes.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing required (5.1 and 5.2).
(c) Quantity of intermediate packages required in shipping container (see 5.2.1).



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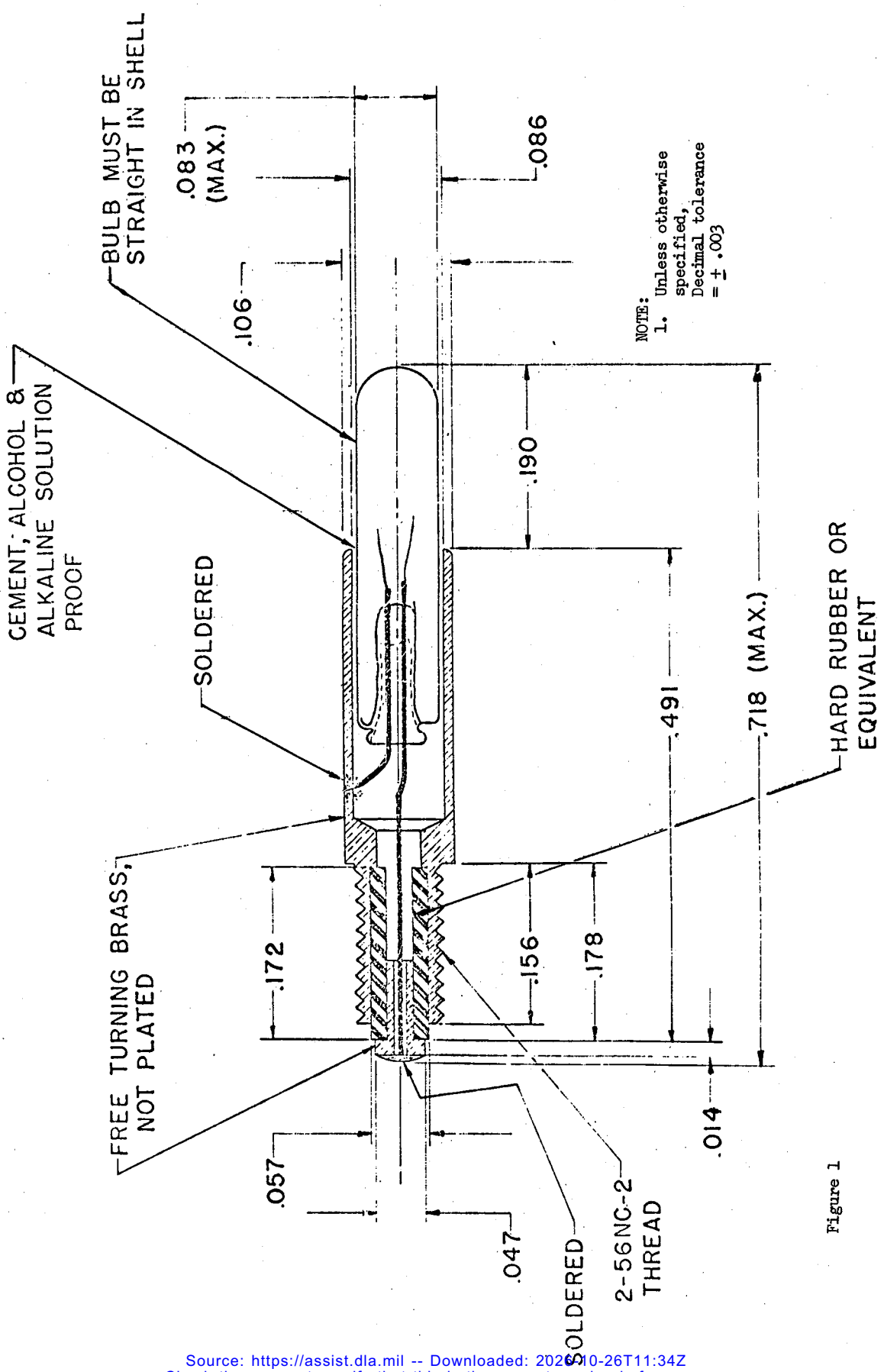



Figure 1

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